

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of NITTE(Deemed to be University)
First Semester M.Tech. (VDE) (CBCS) Degree Examinations
 January 2023

22VDE121 – ADVANCED DIGITAL SYSTEM DESIGN

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.*

Unit – I

Marks BT* CO* PO*

1. a) Draw the block diagram for serial binary adder. Explain the steps involved in writing the state table and hence write the state table. Also draw the State diagram for a serial adder.
- b) Draw the State diagram for a modulo-8 binary counter. Implement modulo-8 binary counter using T flip flop.
2. a) When are two states of a machine said to be k-distinguishable and k-equivalent. Refer to Machine M1 shown in **Figure 2a** and explain how to find the distinguishing sequence for the pair (A, B) and (A, E). Also give an example for equivalent states.

10 L*3 1 3

10 L3 2 3

PS	NS, z	
	x = 0	x = 1
A	E, 0	D, 1
B	F, 0	D, 0
C	E, 0	B, 1
D	F, 0	B, 0
E	C, 0	F, 1
F	B, 0	C, 0

Figure 2a: Machine M1

10 L3 2 3

- b) Consider Machine M1 shown in **Figure 2a**. Give the state partitions and hence write the reduced machine. Explain each step in detail.

10 L2 2 3

3. a) Draw the merger graph and compatibility graph for Machine M2 shown in **Figure 3a**. Hence find the minimal machine.

PS	NS, z			
	I ₁	I ₂	I ₃	I ₄
A	—	—	E, 1	—
B	C, 0	A, 1	B, 0	—
C	C, 0	D, 1	—	A, 0
D	—	E, 1	B, —	—
E	B, 0	—	C, —	B, 0

Figure 3a: Machine M2

10 L3 2 3

- b) With the help of state diagram and state table, design a one-input one-output sequence detector that produces an output value 1 every time the sequence 0101 is detected and an output value 0 at all other times. For example, if the input sequence is 010101 then the corresponding output sequence is 000101

10 L3 1 3

Unit – II

4. a) Draw the map and gate network for the function $T(x,y,z)=\sum(2,3,5,7)$. Explain single input change static hazard. Also explain how to make it hazard free and draw the hazard free network.

10 L3 3 3

- b) Consider the flow table given in **Table 4b**. Write the excitation table and output table. Construct the asynchronous circuit using logic gates by finding the excitation and output functions using K map.

Table 4b: Flow table

	x_1x_2			
	00	01	11	10
A	(A),0	(A),0	(A),0	C
B	(B),1	D	(B),1	C
C	A	(C),1	B	(C),1
D	—	A	—	—

10 L3 3 3

5. a) What static-0 logic hazard and static-1 logic hazard? Explain the static-1 logic hazard using MIC for the circuit with Boolean expression $f = x'y + wx + yz' + wz$. Redesign it to make it Logic-hazard-free network.
- b) What is a race condition in a fundamental model circuit? Comment on critical and non-critical race condition in a fundamental model circuit. Considering the flow table shown in **Table 5b**, write the excitation table and also illustrate the critical race condition for this flow table.

10 L3 3 3

Table 5b: Flow table

	x_1x_2			
	00	01	11	10
A	(A)	C	(A)	C
B	A	(B)	(B)	D
C	D	(C)	B	(C)
D	(D)	B	A	(D)

10 L3 3 3

6. a) Construct the flow table of a sequential circuit that has two inputs x_1x_2 and one output Z; the circuit produces an output of 1 only if the input sequence $x_1x_2 = 10, 11, 01$ is received in that order. All other sequences produce an output of 0.
- b) Explain the dynamic $1 \rightarrow 0$ hazard for the circuit with Boolean expression $f = x'y + wx + yz' + wz + wy$. Redesign it to make it Logic-hazard-free network.

10 L3 3 3

10 L3 3 3

Unit – III

7. a) Define homing tree and distinguishing tree. Consider the state table of a circuit shown in **Table 7a**. Draw the homing tree and distinguishing tree for the state table.

Table 7a: State table

Present state	Input	
	$x = 0$	$x = 1$
A	B,0	D,0
B	A,0	B,0
C	D,1	A,0
D	D,1	C,0

10 L3 4 3

22VDE121

SEE – January 2023

- b) Write the different steps involved in PODEM algorithm. Illustrate the application of PODEM by deriving a test for fault I s-a-1 in the circuit shown in Figure 7b.

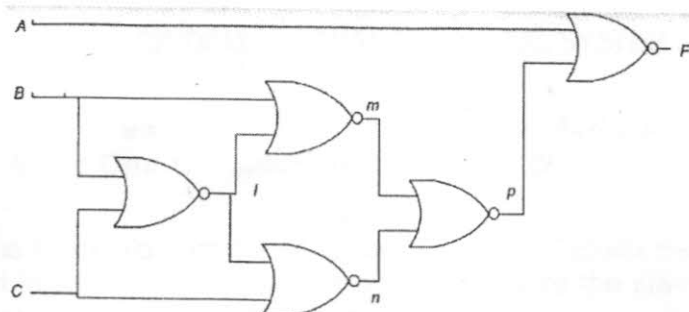


Figure 7b: Circuit under test

10 L3 5 3

8. a) With the help of an example for each, Explain 3 types of cubes used in D algorithm.
b) Illustrate the test generation process by deriving a test sequence for the fault α s-a-0 in the sequential circuit shown in Figure 8b.

10 L2 4 3

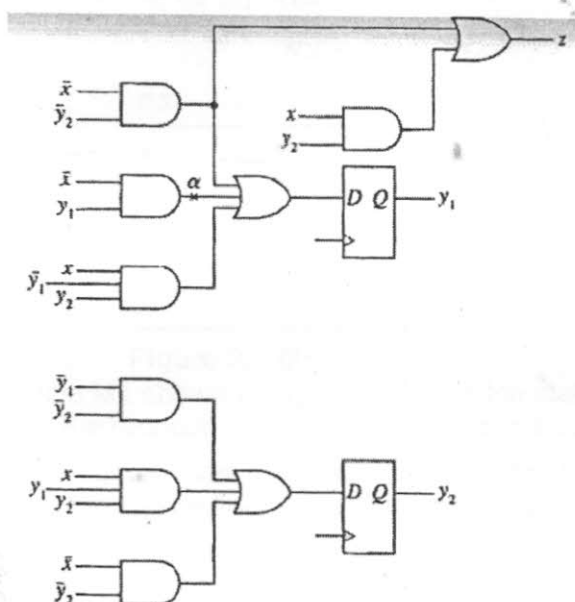


Figure 8b: A sequential circuit

10 L3 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
